

Retail Analytics Dashboard – Business Analysis Report

1. Business Problem Statement

Retail businesses often struggle to clearly understand:

- Which product categories actually drive profit
- Whether discounts are helping or hurting revenue
- Why customers cancel orders
- Whether delivery delays impact performance

The goal of this project is to analyze these areas using data and provide clear, actionable insights for business decision-making.

2. Objective of Analysis

This analysis focuses on:

- Profitability across categories
- Impact of discounting on profit
- Customer cancellation behavior
- Operational delivery performance

The aim is to convert raw transactional data into meaningful business insights using Excel.

3. Dataset Overview

The dataset contains transactional retail data including:

- Category (Furniture, Office Supplies, Technology)
- Sales, Profit, Discount
- Order ID
- Customer Segment (Gender-based grouping)
- Cancellation Flag and Stage
- Delivery Time and Distance

These fields help analyze both financial and operational performance.

4. Business Questions

1. Which category contributes the most profit?
2. How do discounts impact profitability?
3. Which category has the best profit margin?
4. Do customers cancel more before or after shipment?

5. Which customer group has higher cancellation risk?
6. How efficient is the delivery system?
7. Does distance impact delivery delays?

5. Approach Taken

Step 1: Data understanding and cleaning

Step 2: Created new features for better analysis

Step 3: Built pivot tables to summarize data

Step 4: Converted pivots into charts for visualization

Step 5: Designed an interactive dashboard using slicers

The approach focuses on structured analysis followed by visualization.

6. Feature Engineering

To make analysis meaningful, new fields were created:

- Profit % = Profit / Sales
→ Helps measure efficiency, not just total profit
- Cancellation Rate = Cancelled Orders / Total Orders
→ More accurate than raw cancellation count
- Delay Status
→ Categorized orders into Critical, Moderate, On-Time
- Discount Bands
→ Helps identify threshold where profit turns negative

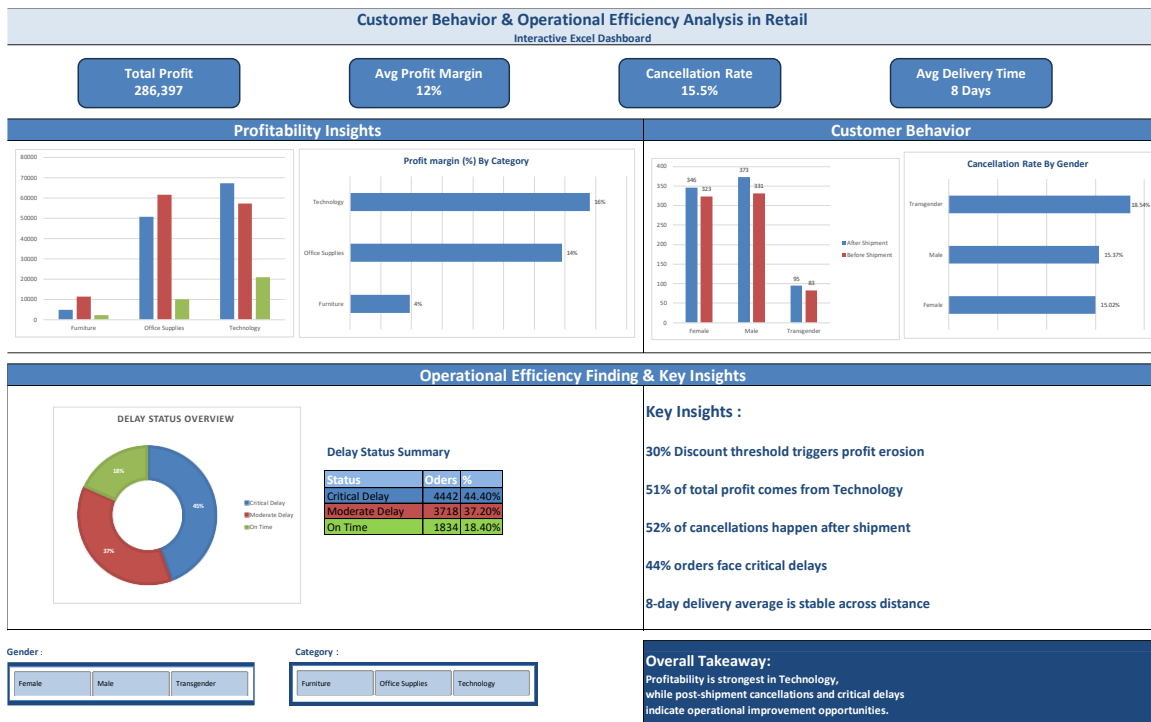
These transformations allowed deeper business-level insights.

7. Key Analysis (Pivot Insights)

- Profit by Category & Gender:
Technology is the highest contributor (~51% of total profit)
- Discount Analysis:
Profit starts declining after ~30% discount
- Profit Margin:
Technology has highest margin (~16%), Furniture lowest (~4%)
- Cancellation Pattern:
52% cancellations occur after shipment

- Cancellation Rate:
Transgender segment has highest cancellation rate (~18.5%)
- Operational Efficiency:
44% orders fall under critical delay category

8. Answers to Business Questions



- Technology is the most profitable category
- High discounts (>30%) negatively impact profit
- Technology has the best profit margin
- More cancellations happen after shipment
- Certain customer groups show higher cancellation risk
- Delivery performance shows high delay percentage
- Distance has minimal impact on delivery time

9. Key Business Insights

- Discount threshold is a major risk factor for profitability
- Profit is heavily concentrated in Technology category
- Post-shipment cancellations indicate operational inefficiency
- High percentage of delays suggests process issues
- Delivery time consistency indicates internal system bottlenecks

10. Conclusion

This project demonstrates how structured analysis using Excel can convert raw data into business insights.

The dashboard provides:

- Clear performance tracking
- Data-driven decision support
- Interactive filtering for deeper exploration

It reflects strong analytical thinking, problem-solving approach, and ability to communicate insights effectively.